

Pictures Become Numbers

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Rules or Learned Patterns?

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Input, Process, Output

Without opening the old lesson, explain one key idea from Input, Process, Output. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Preserve the missing field for Images as numerical grids. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Images as numerical grids. Proposed explanation: “The number one always means black”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Design a new counterexample

Create two different images of width 4 and height 2, each with exactly three black pixels. Write both complete row encodings. Explain which fact their shared count retains and which difference it discards.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first. Explain why the image changes although the six values are identical.

6. Solve the given case

A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

Draw it, then change the top-left value to 1.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Use 0 for dark and 1 for light. Draw a two-by-two image with rows [1,0] and [0,1]. Then change only the bottom-left pixel to light. Which number changes?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 5 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Choose which visible item features the Classroom Sorter may represent as image measurements and name one feature that lighting could distort.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first. Explain why the image changes although the six values are identical.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Read and reconstruct

Using 1 for black and 0 for white, draw width 3, height 2 with rows 1,0,1 and 0,1,0. Read it back into numbers and count black pixels.

My reasoning and evidence

13. W2 Preserve the missing field

A message contains 1,0,0,1,1,0 but gives no dimensions. Show a one-row arrangement and a two-row arrangement using the same values. Name the extra information that would distinguish them.

My reasoning and evidence

14. W3 Test a count feature

A classifier receives only the number of black pixels. Can it always distinguish the chapter's cross from every other three by three pattern? Construct a different pattern with the same count and explain.

My reasoning and evidence

15. W4 Design a new counterexample

Create two different images of width 4 and height 2, each with exactly three black pixels. Write both complete row encodings. Explain which fact their shared count retains and which difference it discards.

My reasoning and evidence

16. W5 Locate an input problem

A program receives the correct grid but displays black wherever the code says 0 and white wherever it says 1. A student proposes collecting more classifier examples. Identify the earlier mismatch and the first check to perform.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Pictures Become Numbers

Two fictional 2-by-3 pixel badges contain the same number of dark squares.

Dark=1, light=0; rows are read top to bottom, left to right.

Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

Pixel positions: two badges, same dark count

	Left	Middle	Right
A top	1	0	1
A bottom	0	1	0
B top	1	1	1
B bottom	0	0	0

Read a cell using both its row and column labels.

Pictures Become Numbers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Pixel convention

Each image has two rows and three columns; 1 means dark.



2. Badge A

Top 101, bottom 010: three dark pixels.



3. Badge B

Top 111, bottom 000: also three dark pixels.



4. Compare arrangements

The totals agree, but middle-top and middle-bottom positions differ.

CHANGE ONE THING

Change A's bottom-left pixel from 0 to 1.

Explain the changed result using the matching lesson.

Find a position where the original badges differ, then read the changed A back as a picture.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-pixel count describes shape.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-pixel count describes shape.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Two fictional 2-by-3 pixel badges contain the same number of dark squares.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The totals agree, but middle-top and middle-bottom positions differ.

1. Compare arrangements
2. Badge B
3. Pixel convention
4. Badge A

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare arrangements The totals agree, but middle-top and middle-bottom positions differ. A grid preserves where each brightness belongs. Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges. Dimensions and reading order belong to the representation, not to an optional caption.

Practice 2: explain the deciding evidence

Changed-case practice: Change A's bottom-left pixel from 0 to 1. Which conclusion is supported by the supplied facts?

1. A now has four dark pixels; its row encoding is 101 / 110.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: A now has four dark pixels; its row encoding is 101 / 110. Exactly one light pixel becomes dark; the other five positions remain unchanged.

Repair a missing building block

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used labels and which stage used only the new measurement.. Return to this week and explain what you changed.

Review Input, Process, Output (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
